

Interview Signals

A structured-interview platform with a behavioural measurement layer. The interview engine is **ready to pilot**. The measurement layer is built and tested but **must not touch a hiring decision** until a fairness audit and a consent notice exist.

VERSION	REVIEWED	TEST SUITES	PARAMETERS
v1.1 · 31 Aug 2026	2 Sep 2026	15, all passing	157 catalogued
JURISDICTION			
India · DPDP 2023			

READY TO PILOT 1 The structured-interview engine. Questions, anchored ratings, independent locking, audit trail. Usable now, with no measurement enabled.	BLOCKING A REAL HIRE 3 No interview consent notice, no fairness audit, and no signed processor agreement for question generation. All three need a decision above engineering.	NEVER TO BE BUILT 4 Face search, emotion inference, a hireability score, and any automated decision. Excluded by design, not yet on a roadmap.
RECURRING COST 1 One paid dependency, for drafting interview questions from a CV. Everything else runs on the interviewer's own machine.		

What this is

Two things, deliberately kept apart.

The decision layer is a structured interview. Every candidate for a role is asked the same questions in the same order. Interviewers score against written behavioural anchors — concrete descriptions of what a 1 and a 5 look like — quote the evidence behind each score, and lock their ratings before seeing anyone else's. That sequence is the product. It works with no camera at all.

The measurement layer reports what a face and body did. From a webcam it extracts facial action-unit proxies, upper-body posture and gesture, gaze and blink dynamics, contactless pulse, voice prosody and transcript content — as time-aligned numbers, once a second.

It does not output a hireability score, a personality profile, a truthfulness estimate, or an emotional state presented as fact. Not because those are hard, but because none of them can be validly derived from these signals, and shipping them is what turns a defensible product into an indefensible one.

Structured interviews are the strongest single predictor of job performance available. Facial and physiological signals have no established validity against job performance at all.

SACKETT ET AL. 2022 · RE-ANALYSIS OF THE PERSONNEL-
SELECTION LITERATURE

OPERATIONAL VALIDITY AGAINST JOB PERFORMANCE		
Predictor	r	Role in this product
Structured interview	.42	Built. This is the decision layer.
Job knowledge test	.40	Not built. Cheap, strong complement.
Empirically keyed biodata	.38	Not built. Needs historical outcome data.
Work sample	.33	Not built. Highest candidate acceptance.
Cognitive ability	.31	Not built. Carries adverse-impact exposure.
Unstructured interview	.19	What most hiring in India does today.
Facial / physiological signals	—	Built as measurement only. Not an input to any score.
The commercial argument in one line: the structured-interview engine alone more than doubles the predictive validity of what most employers currently do, and it carries none of the legal exposure of the signal layer.		

What has been built

Delivered as numbered work packages, in dependency order. Each was completed with tests that fail if the behaviour regresses.

WP0

Tunable configuration with an auditable identity

IMPLEMENTED

Every threshold in the pipeline lives in one file and hashes to a short digest written into each session's output. That is what lets you prove two recordings were measured with the same instrument — or prove they were not.

WP0b

Property tests for the silent-failure classes

IMPLEMENTED

The failures that matter here do not raise an error; they return a plausible wrong number. These tests target that class specifically.

WP1	Facial and upper-body signal extraction 38 action-unit proxies, head pose, gaze, blink dynamics, shoulder line, lean, sway, gesture energy, self-touch.	IMPLEMENTED
WP2	Contactless pulse and voice prosody Pulse from video colour change, using the published POS algorithm with a signal quality index. Voice measures computed with Praat's own algorithms so they are comparable to the phonetics literature.	IMPLEMENTED
WP3	Structured interview engine Guides with competencies and anchored scales, fixed question order, independent rating, locking before reveal, amendments that preserve the original, and a full audit trail. Rejects a guide whose competencies are impressions rather than behaviours — "culture fit", "confidence", "presence".	IMPLEMENTED
WP4	On-device transcription and content measures Speech to text runs on the interviewer's machine. No candidate speech reaches a hosted transcription service.	IMPLEMENTED
WP5	The interview platform, end to end Scheduled invitations, a candidate call interface, an interviewer instrument panel, speaker-attributed live transcript, two-way video, and a regional gate that disables measurement without disabling the interview.	IMPLEMENTED
WP7a	Consent record and withdrawal machinery Itemised consent naming each signal, enforced at the capture path rather than in the interface. Withdrawal deletes everything and completes in seconds. Retention expires automatically.	IMPLEMENTED
WP7b	CV-derived interview questions Upload a candidate's CV and get extra follow-up questions grounded in what it claims, at a difficulty set for the role. Also suggests follow-ups from what the candidate has just said. Nothing generated is ever scored.	IMPLEMENTED

Measuring how accuracy varies across skin tone and other subgroups. Not started. This is the first item in the release gate below.

Identity: what the camera can and cannot tell you

This is worth stating precisely, because the two things get confused and the difference is the whole legal position.

What is built is a closed-set match against an enrolled, consenting gallery. A colleague runs an enrolment command against their own consent record, which stores one face template under their own subject folder on that machine. The camera can then answer one question: *which of the people who have enrolled here is in front of it, if any?* Five people enrolled means five people it can name. It resolves them to their record — prior sessions, what they consented to, and the professional links they attached themselves — so a recording lands against the right person.

What is not built, and is structurally prevented, is face search. There is no index of the web, no scraped database, no third-party lookup, and no code path that takes a face and asks anyone else who it belongs to. An unenrolled face returns UNKNOWN by construction rather than by policy: a template only exists because a person ran enrolment against their own consent, so there is nothing to match a stranger against.

That guarantee is load-bearing. Open-ended face search against candidates is precisely the biometric-identification use the EU AI Act restricts, and under DPDP 2023 it has no lawful basis in hiring — there is no consent from the person being searched for, and no purpose limitation that would survive contact with a regulator. The moment the gallery is populated from anywhere but explicit enrolment, this becomes a different system with a different legal position and a different answer to every question below.

Deliberately excluded

These are the questions the product will be asked to do, and the reasons it does not. Worth reading before promising any of them.

Face search against external databases

EXCLUDED

Taking a candidate's face and finding out who they are, or what else is known about them, from any source outside this machine.

Why not: no lawful basis under DPDP 2023, restricted under the EU AI Act, and it would destroy the enrolled-gallery guarantee that makes the identity feature defensible. The face resolves *who*; the person supplies *what*.

Emotion or state inference

EXCLUDED

Reporting that a candidate was nervous, dishonest, engaged or confident.

Why not: EU AI Act Article 5(1)(f) prohibits inferring emotions from biometric data in workplace contexts, and the guidelines confirm hiring falls inside that. India has no equivalent statute today — that is a gap in the law, not a licence, and building the core of a product on that gap staying open is a poor bet.

A hireability score

EXCLUDED

Any single number ranking candidates, derived wholly or partly from the signals.

Why not: the signals have no demonstrated predictive validity against job performance. A score built on them would look authoritative and mean nothing, which is worse than no score.

Automated decisions

EXCLUDED

The system advancing or rejecting a candidate, or ordering a shortlist.

Why not: a human makes every decision and records their reasoning against the anchors. The measurement output is not wired to the rating interface at all, and that separation is enforced in code and asserted by tests.

Scope, in numbers

The parameter catalogue is generated from the pipeline source, so the scope is a count rather than a claim. Each parameter carries a validity tier stating how far it can be trusted.

157 PARAMETERS BY GROUP AND BUILD STATUS			
Group	Total	Core set	Status
A · Facial action units	102	2	IMPLEMENTED
B · Upper-body kinesics	11	1	IMPLEMENTED
C · Physiological (pulse)	8	3	IMPLEMENTED
D · Vocal prosody	19	12	IMPLEMENTED
E · Linguistic content	7	5	IMPLEMENTED
F · Session integrity	10	7	IMPLEMENTED
146 implemented · 6 deferred to Phase 1 · 2 to Phase 2 · 3 excluded outright. The 30-parameter core set is what a first deployment would actually report; the rest exist so the catalogue is complete and auditable.			

VALIDITY TIER — HOW FAR EACH PARAMETER CAN BE TRUSTED		
Tier	Count	Meaning
Measured	131	A physical quantity the system observes directly. Reportable as fact.
Weak inference	23	Reportable only with its caveat attached. Never on its own.
Do not ship	3	Catalogued so the exclusion is explicit and reviewable. Not built.

What the tests actually prove

Fifteen suites run on every change and all pass. The ones worth quoting to a customer are the ones that prove the system refuses to report a number it cannot support.

VERIFIED BEHAVIOUR	
Claim	Evidence
Pulse measurement is accurate	Recovers synthetic pulse rates from 48–130 BPM to under 0.2 BPM, and holds under 0.5 BPM through simulated head motion.
...and refuses when it cannot be	Under heavy noise the quality index drops and the estimate is discarded rather than displayed. Flat regions and short buffers return nothing, never a fabricated number.
Blink detection does not invent blinks	Counts 0 blinks across 300 frames of threshold chatter. A naive single-threshold detector reports about 150.
A camera switched off stops the measurement	When a candidate turns their camera off, the panel blanks and the rolling measurement state is discarded — so a camera coming back is never measured against samples from before the gap. Without that, the pulse estimator reported a two-minute-old reading as current.
Generated questions can never become a score	Ratings are made against the guide's competencies only. CV-derived questions carry no score by construction, and the test asserts none can appear as a rated row.
Identity refuses strangers and near-ties	An unenrolled face returns UNKNOWN. Where the top two candidates are close, the result is refused rather than guessed, and a human types the name.
Ratings stay independent	No interviewer sees another's scores until everyone has locked. A single interviewer can run an interview end to end without that constraint blocking them.

Legal position

India, where this deploys. The Digital Personal Data Protection Act 2023 is operative, with Rules notified on 14 November 2025. Facial imagery, biometric features and physiological measurements are all personal data under it. That requires informed, specific, freely given consent with an itemised notice; purpose limitation; retention limits with real deletion; a working withdrawal path; and breach notification. Penalties reach ₹250 crore per contravention.

The machinery for all of that is built and enforced in code. What does not yet exist is the notice itself — see the release gate below.

One trap worth naming. "Consent or forfeit the interview" is coercive by construction and the consent will not hold. Every consent-gated feature in the build is therefore refusable without disadvantage: a candidate who declines is interviewed on the same fixed questions, and refusal is not recorded as anything about them.

Exposure that may not be priced in. EU AI Act Article 5(1)(f) prohibits emotion inference from biometric data in workplace contexts, and applies extraterritorially to systems whose output is used in the EU. If a client ever hires into Europe with this, that provision reaches the product regardless of where it runs. The exclusions above are what keep it out of scope.

Release gate

Ordered by what blocks first. The first three are decisions above engineering — they need budget, counsel, or a signature, not code.

GATE 1 **There is no interview consent notice** BLOCKER

The notice that exists covers *research recording sessions*. It tells the reader "you are not applying for a job with us", and ends by saying that using these signals in a real interview is a different consent problem which it does not address. That assessment is correct and the interview notice has not been written.

The candidate interface currently shows candidates that research notice, which is the wrong document. Until the right one exists and counsel has signed it, no real candidate should be measured.

NEEDS:

legal counsel, and a decision on who owns the notice. Engineering cannot write it.

GATE 2 The fairness audit has not been run BLOCKER

Pulse-from-video error and action-unit detection accuracy both vary with skin tone. That is a known property of the underlying methods, not a defect in this build. Unaudited, that variance becomes discrimination the moment the output influences a hiring outcome — and it would be invisible in aggregate results.

The recruitment brief for the validation cohort is written. The study has not been run, so the size of the variance in this system is currently unknown.

NEEDS:

budget and time for a recruited, paid, consented cohort with contact-sensor ground truth. This is the single largest outstanding item.

GATE 3 CV question generation sends data to a third party NEEDS SIGN-OFF

Every other part of this system processes data on the interviewer's own machine, by design. The CV-question feature is the single exception: it sends the CV text and the candidate's answer transcript to an external model provider to draft questions. Contact details are stripped first; name, employers and city are not, because the questions are about that experience.

The processing record and the draft candidate disclosure are written. What is missing is a data processing agreement, a statement of the provider's retention terms, and counsel's view on cross-border transfer. Until then the feature can be switched off with one setting, and the interview is unaffected.

NEEDS:

a signed DPA and counsel's sign-off on the disclosure text.

COST:

roughly ₹2-8 per candidate at current pricing.

GATE 4 **Question difficulty must be set per role, not per candidate**

POLICY NEEDED

CV-derived questions are generated at one of four difficulty bands. If the band is chosen per candidate — especially after reading their CV — then an interviewer's impression of someone sets the difficulty they face, and difficulty sets their score. That is the exact mechanism structured interviewing exists to remove.

The build records the band on every interview, refuses to change it once questions have been generated, and warns in the summary when one candidate was probed differently from others for the same role. It cannot prevent divergence, only make it visible.

NEEDS:

a written policy that the band is fixed per requisition before candidates are seen. Optionally, we block divergence outright.

GATE 5 **Do not connect the measurement output to a decision**

STANDING RULE

Not an outstanding task — a rule that has to survive the next several product conversations. The measurement layer is descriptive. The moment a number from it is wired into a rating, a ranking or a shortlist, every exclusion above stops being true and the legal analysis restarts.

Its legitimate use is live capture quality: telling an interviewer that the candidate is backlit, out of frame or dropping frames, while there is still time to fix it.

NEEDS:

agreement that this holds, including when a customer asks for a score.

What it costs to run

DEPENDENCIES AND RUNNING COST		
Item	Where it runs	Cost
Face, pose and pulse measurement	Interviewer's machine	None. Model weights ship with the build.
Speech to text	Interviewer's machine	None. Chosen over a hosted service so candidate speech stays local.
Identity match	Interviewer's machine	None. Optional model, no network path.
Interview engine and platform	Your own server	Hosting only. No third-party service.
CV question generation	External provider	Roughly ₹2-8 per candidate. The only paid dependency, and removable.
Deployment note: browsers block camera and microphone access on plain HTTP except on localhost, so this needs TLS in front of it before any real candidate joins.		

Recommendation

Pilot the structured-interview engine now, with measurement switched off. It is the part with demonstrated validity, it carries none of the biometric exposure, and it is a saleable product on its own in a market where most interviewing is unstructured. Nothing in the release gate blocks it.

Fund the fairness audit as a separate, explicit decision. It is the long pole and it gates every use of the measurement layer. Until it is done, we do not know the size of the variance we would be shipping, and "we did not measure it" is not a position that survives a complaint.

Treat the exclusions as product strategy, not as engineering caution. Every competitor claim about detecting confidence, honesty or emotion from a face is a claim we can decline to make and can explain why. That is a sales asset in front of a client's legal team, and it becomes worthless the moment we quietly ship a score.

Interview Signals v1.1 · review dated 2 September 2026 · 15 test suites passing

Source documents in the repository: spec.html (evidence and legal analysis), docs/WP7a (research consent notice), docs/WP7b (question-generation processing record), docs/WP8a (fairness-audit recruitment brief), out/interview-parameters.xlsx (parameter catalogue).

Figures for validity are from Sackett et al. 2022. Legal statements summarise the analysis in the specification and are not legal advice.